

Prob 1 - Rounding errors

$$\left| \frac{u(x-h) - 2u(x) + u(x+h))}{h^2} - u''(x) \right| = O(h^2)$$

$$O(\epsilon/h^2) = O(h^2) \Rightarrow O(\epsilon/h^2) = O(h^2) \Rightarrow O(\epsilon) = h^4 \Rightarrow h = \sqrt[4]{\epsilon} = \sqrt[4]{10^{-16}} = 10^{-4}$$

Prob 3 - Fixed-point method

$$f(x) = 2x^3 - x^2 + 2x - 1 = 0$$

$$x = \frac{-2x^3 + x^2 + 1}{2} = g_1(x)$$

$$x = \sqrt[3]{\frac{x^2 - 2x + 1}{2}} = g_2(x)$$

$$g_1'(x) = -3x^2 + x$$

$$g_2'(x) = \frac{1}{3} \cdot \left(\frac{x^2 - 2x + 1}{2} \right)^{\frac{1}{3}-1} (2x - 1) = \frac{x-1}{3 \left(\frac{x^2 - 2x + 1}{2} \right)^{\frac{2}{3}}}$$

Prob 4 - Fixed-point iterations

$$a) \quad g(x) = -1,93 \ln(1,32x/5000) \Rightarrow g'(x) = -1,93 \cdot \frac{1,32}{5000} \cdot \frac{5000}{1,32x} = -\frac{1,93}{x}$$

$g'(x)$ is decreasing $\Rightarrow g(x)$ is a decreasing function

b) Since we know that $g(x)$ is a decreasing function, the max and min values is at the

$$g_{\max}(e) = -1,93 \ln(1,32e/5000) = 13,97$$

$$g_{\min}(e^3) = -1,93 \ln(1,32e^3/5000) = 10,11$$

$$c) \quad g'(x) \text{ is biggest when } x \text{ is smallest} \Rightarrow g'_{\max}(e) = -\frac{1,93}{e} = -0,71 \Rightarrow L = 0,71$$

$$d) \quad x^{k+1} = g(x^k), \quad x^0 = e^2$$

$$x^1 = g(x^0) = g(e^2) = -1,93 \ln(1,32e^2/5000) = -12,04$$

$$e) \quad |x_{k+1} - r| \leq \frac{L}{1} |g(x_0) - x_0|$$

$$10^{-3} = \frac{0,71}{0,71} | -12,04 - e^2 |$$

$$10^{-3} = 0,71^k \cdot 19,43$$

$$\frac{1}{0,01943} = 0,71^k \Rightarrow k = \frac{\ln(0,01943)}{\ln(0,71)} = -11,5 \Rightarrow k = 12$$